

- 3 (a) (i) Molecules collide with one another in a haphazard manner hence they possess different kinetic energies at any time. B1
(word "random" does not get marks in answer)
- (ii) No intermolecular forces in ideal gas, hence no potential energy B1
(Do not accept specific forces eg. "attraction"; vague description eg. "interaction")
- (iii) $p = 1/3 \rho \langle c^2 \rangle$ M1
 $\Rightarrow pV = 1/3 M \langle c^2 \rangle$ since $\rho = M/V$
For ideal gas, $pV = nRT$
Hence, $1/3 M \langle c^2 \rangle = nRT$ B1
Total KE = $\frac{1}{2} M \langle c^2 \rangle = \frac{3}{2} nRT$
(Must start with "Use the equation" given, other methods not accepted)
- (iv) Low pressure B2
High temperature
(Low pressure implies molecules are far apart, high temperature implies high KE)
- (b) (i) $pV = nRT$ M1
 $(6.0 \times 10^5) (0.10) = (10) (8.31) T$
A1
 $T = 722 \text{ K}$
- (ii) The products of p and V are not constant eg. the product of p and V is $5.0 \times 10^4 \text{ J}$ at B, but $6.0 \times 10^4 \text{ J}$ at A. B1
(or calculate temperature at B = 600 K, lower than 722 K at A)
- (iii) Work done in one cycle estimated from the area ABC enclosed M1
 $= 8.8 \times 10^4 \text{ J}$
A1
*(Note positive sign since net work is done by the gas
Do not accept area "under curve")*

(iv) In one cycle $\Delta U = q + W = 0 \Rightarrow q = 8.8 \times 10^4 \text{ J}$
(*working or otherwise clear explanation required*)

A1

4. (a) (i) 1 cycle